

 Sae-A Trading Co., Ltd.		Fleece & Brushed Fabric Quality Control		Doc. No. : EN65
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Approved by	Connie Park	Date Validation	Until New Advise	
Objective	The purpose of this SOP is to establish standardized control requirements for managing the key quality risks associated with Fleece and Brushed Fabric products. By standardizing all processes from fabric receiving to final shipment, this SOP aims to ensure consistent product quality that meets customer requirements, prevent consumer complaints, and improve customer satisfaction.			
Note : If current SOP content differs from Sae-a Manual or data, this SOP takes precedence. The following SOP may include but not limited to Sae-a minimum requirements. When it comes to circumstances not described in Sae-a SOP, follow Sae-a HQ/Local QA Team's guidelines. Factory's SOP must meet or exceed Sae-a SOP requirements.				
Description of Activities				
No.	Section/ Responsible	Activities	Documentation	
1		Scope This SOP applies to all fabrics and garments made from the following brushed fabrics: • Fleece • Polar Fleece • Micro Fleece • Sherpa Fleece • Other brushed fabrics		
2	Fabric Warehouse Chief/Manager	Fabric Storage 2.1 Fabric rolls shall be stacked no higher than max. 3 rolls to prevent pile compression and appearance deformation. 2.2 Maintain plastic covers and store fabrics in a clean environment to protect them from moisture and dust contamination.		
3	Fabric Inspection QC	Fabric Inspection Requirements 3.1 Fabric Width Verification • Measure cuttable fabric width. • Check for selvedge curling. Since brushed fabrics tend to curl at the edges, cuttable width shall be measured based on the usable fabric width. 3.2 Thickness & Hand-feel Verification • Verify thickness uniformity (Uneven Thickness) • Verify hand-feel consistency (Hard/Soft Hand-feel) Check that brushing is evenly distributed and that there are no localized areas with excessive or insufficient brushing. 3.3 Brushing Quality Verification • Brushing lines / brushing marks • Excessive or insufficient brushing • Exposed ground fabric • Missing or damaged pile • For fabrics brushed on the inside, inspect the brushing quality on the inside surface. 3.4 Hole & Needle Damage Inspection Since pile may conceal defects such as holes, needle damage, and broken yarns, the use of the inspection machine's backlight is strongly recommended during fabric inspection. 3.5 Shade Verification • Verify shade with the pile direction aligned in the same direction. • Check roll-to-roll shade variation • Check side-to-center and end-to-end shade variation within each roll.		
4	In-house Lab Chief/Manager	In-house Lab Test Requirement 4.1 Fabric Weight • Verify GSM variation caused by brushing inconsistency. 4.2 Pilling Test (if applicable) 4.3 Appearance After Laundering • Pile Damage • Missing Pile • Surface Distortion		

		- Shedding Test Method a. Wear white or contrast color gloves. b. Lay the fabric flat. c. Rub the brushed surface 10 times using the entire palm with consistent pressure. d. Evaluate the amount of loose fibers attached to the glove. **Accept Criteria** <table><thead><tr><th>Grade</th><th>Condition</th><th>Result</th></tr></thead><tbody><tr><td>Grade 5</td><td>No fiber attachment</td><td>Accept</td></tr><tr><td>Grade 4</td><td>Slight fiber attachment</td><td>Accept</td></tr><tr><td>Grade 3</td><td>Visible fiber attachment</td><td>Marginal Accept</td></tr><tr><td>Grade 2</td><td>Significant fiber attachment</td><td>Reject</td></tr><tr><td>Grade 1</td><td>Severe fiber shedding</td><td>Reject</td></tr></tbody></table> 4.4 Wear & Wash Test - Evaluate shedding, fiber transfer, pile loss, and overall appearance after actual wear and home laundering. - Detailed test procedures and acceptance criteria shall follow the "Sae-A Wear & Wash Test SOP".	Grade	Condition	Result	Grade 5	No fiber attachment	Accept	Grade 4	Slight fiber attachment	Accept	Grade 3	Visible fiber attachment	Marginal Accept	Grade 2	Significant fiber attachment	Reject	Grade 1	Severe fiber shedding	Reject	Wear & Wash Test SOP Doc. No.: KR54 EN54
Grade	Condition	Result																			
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Grade 1	Severe fiber shedding	Reject																			
5	Cutting Manager	Cutting Control 5.1 Fabric Relaxation - Fabric shall be relaxed for a minimum of 24 hours prior to cutting. A relaxation period of 48 hours is recommended. 5.2 Spreading Control - Do not mix different fabric lots or shade lots during spreading. 5.3 Spreading Height Control - Fleece: Maximum lay height of 7 inches. More stringent lay height limits may be applied depending on fabric characteristics and garment style. 5.4 One-Way Cutting - One-way cutting shall be applied to prevent shade variation and appearance differences caused by inconsistent pile direction. 5.5 Bundle Control - Cut bundles shall be stored at an appropriate height to prevent pile compression and deformation.																			
6	Sewing, Finishing & Packing Chief/Manager	Sewing, Finishing & Packing Control 6.1 Trimming & Fiber Removal - After sewing, remove loose fibers and lint using one or more of the following: - Air Blower - Vacuum Cleaner - Roller Type Lint Remover 6.2 Ironing / Finishing - Use steam ironing - Steam temperature shall be validated and controlled to prevent pile damage and compression caused by excessive heat.																			
		6.3 Metal Detection - Since brushing processes may leave broken brushing machine wires or metal fragments inside the fabric, all fleece and brushed fabric garments shall undergo 200% metal detection, regardless of buyer requirements. 6.4 Packing Control - Vacuum packing or compressed packing is prohibited to prevent pile compression, appearance damage, and garment deformation. 6.5 Long-Term Storage & Shipment - Polyester fleece fabrics tend to retain moisture within their pile structure. For long-term storage or shipment, Super Dry desiccants shall be placed inside the polybag and carton to minimize the risk of mold formation.																			
7	QC/QA	Inline & Pre-final Inspection 7.1 Shedding Test - Rub the garment surface repeatedly using a white glove or contrasting fabric. - Acceptance criteria shall follow Section 4.3 (Shedding Acceptance Criteria).																			

		7.2 Visual Inspection • Brushing Quality • Shade Variation • Pile Direction Consistency • Missing/Damaged Pile • Press Mark • Contamination	
8		Remediation (Shedding Test Failure) • If the garment fails the Shedding Test, first remove loose fibers using air blowing, vacuum cleaning, or a lint roller. • If the issue cannot be sufficiently improved, consult with the relevant departments (QA, Production, Technical, Production Planning, and Merchandising) to evaluate the application of garment washing. • Since garment washing may result in additional quality risks, the following items shall be evaluated before implementation: - Measurement Change - Skew/Torque - Appearance Change - Pile damage or brushing changes - Shade Variation • Prior to bulk production, pilot testing and validation shall be conducted to confirm that no unacceptable quality issues are identified. Production shall proceed only after final approval.	